5-1g Other Demand Elasticities

In addition to the price elasticity of demand, economists use other elasticities to describe the behavior of buyers in a market.

The Income Elasticity of Demand

The

income elasticity of demand (a measure of how much the quantity demanded of a good responds to a change in consumers' income, computed as the percentage change in quantity demanded divided by the percentage change in income)

measures how the quantity demanded changes as consumer income changes. It is calculated as the percentage change in quantity demanded divided by the percentage change in income. That is,

$$\text{Income elasticity of demand} = \frac{\text{Percentage change in quantity demanded}}{\text{Percentage change in income}}.$$

As we discussed in Chapter 4, most goods are *normal goods*: Higher income raises the quantity demanded. Because quantity demanded and income move in the same direction, normal goods have positive income elasticities. A few goods, such as bus rides, are *inferior goods*: Higher income lowers the quantity demanded. Because quantity demanded and income move in opposite directions, inferior goods have negative income elasticities.

Even among normal goods, income elasticities vary substantially in size. Necessities such as food tend to have small income elasticities because consumers choose to buy some of these goods even when their incomes are low. Indeed, a long-established empirical regularity is *Engel's Law* (named after the statistician who discovered it): As a family's income rises, the percent of its income spent on food declines, indicating an income elasticity less than one. By contrast, luxuries such as jewelry and recreational goods tend to have large income elasticities because consumers feel that they can do without these goods altogether if their incomes are too low.

The Cross-Price Elasticity of Demand

The

cross-price elasticity of demand (a measure of how much the quantity demanded of one good responds to a change in the price of another good, computed as the percentage change in quantity demanded of the first good divided by the percentage change in price of the second good)

measures how the quantity demanded of one good responds to a change in the price of another good. It is calculated as the percentage change in quantity demanded of good one divided by the percentage change in the price of good two. That is,

$$\text{Cross-price elasticity of demand} = \frac{\text{Percentage change in quantity demanded of good one}}{\text{Percentage change in the price of good two}}$$

Whether the cross-price elasticity is positive or negative depends on whether the two goods are substitutes or complements. As we discussed in Chapter 4, *substitutes* are goods that are typically used in place of one another, such as hamburgers and hot dogs. An increase in hot dog prices induces people to grill more hamburgers instead. Because the price of hot dogs and the quantity of hamburgers demanded move in the same direction, the cross-price elasticity is positive. Conversely, *complements* are goods that are typically used together, such as computers and software. In this case, the cross-price elasticity is negative, indicating that an increase in the price of computers reduces the quantity of software demanded.

QuickQuiz

1. A good tends to have a small price elasticity of demand if
   a. the good is a necessity.
   b. there are many close substitutes.
   c. the market is narrowly defined.
   d. the long-run response is being measured.

2. An increase in a good's price reduces the total amount consumers spend on the good if the ____ elasticity of demand is ____ than one.
   a. income; less
   b. income; greater
   c. price; less
   d. price; greater

3. A linear, downward-sloping demand curve is
   a. inelastic.
   b. unit elastic.
   c. elastic.
   d. inelastic at some points, and elastic at others.

4. The citizens of Lilliput spend a higher fraction of their income on food than do the citizens of Brobdingnag. The reason could be that

    a. Lilliput has lower food prices, and the price elasticity of demand is zero.

    b. Lilliput has lower food prices, and the price elasticity of demand is 0.5.

    c. Lilliput has lower income, and the income elasticity of demand is 0.5.

    d. Lilliput has lower income, and the income elasticity of demand is 1.5.

Chapter 5: Elasticity and Its Application: 5-1g Other Demand Elasticities
Book Title: WGU Economics
Printed By: Mike Benton (mbento6@wgu.edu)